18. Three Entrances A study was conducted to aid in the remodeling of an office building that contains three entrances. The choice of entrance was recorded for a sample of 200 persons who entered the building. Do the data in the table indicate that there is a difference in preference for the three entrances? Find a 95% confidence interval for the proportion of persons favoring entrance 1.

Entrance	1	2	3
Number Entering	83	61	56

$\boxed{p_1 = p_2 = p_3 = \frac{1}{3}}$

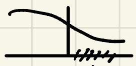
♡ 1
♡ 1 ∴ 二項分佈

① $\begin{cases} H_0: p_1 = p_2 = p_3 = \frac{1}{3} \\ H_a: \exists\, p_i \neq \frac{1}{3} \end{cases}$ $O_1 = 83 \quad O_2 = 61 \quad O_3 = 56$

$E_i = 200 \times \frac{1}{3} = 66.67$

$\therefore \chi^2 = \frac{(83-66.67)^2}{66.67} + \frac{(61-66.67)^2}{66.67} + \frac{(56-66.67)^2}{66.67} = 6.19$

$df = k-1 = 3-1 = 2 \quad \alpha = 0.05 \rightarrow \chi^2_{2,0.05} = 5.991$



5.991

∴ reject H_0 ∴ 存在偏好差異

② $CI = \hat{p} \pm z_{0.025} \cdot \sqrt{\frac{\hat{p}(1-\hat{p})}{n}} = (0.3468, 0.4832)$ #

22. Heart Attacks on Mondays Researchers from Germany have concluded that the risk of a heart attack for a working person may be as much as 50% greater on Monday than on any other day.[1] In an attempt to verify their claim, 200 working people who had recently had heart attacks were surveyed and the day on which their heart attacks occurred was recorded:

Day	Observed Count
Sunday	24
Monday	36
Tuesday	27
Wednesday	26
Thursday	32
Friday	26
Saturday	29

$n = 200$

$k = 7$

$\alpha = 0.05$

Do the data present sufficient evidence to indicate that there is a difference in the incidence of heart attacks depending on the day of the week? Test using $\alpha = .05$. → no.

$\begin{cases} H_0: p_1 = \ldots = p_7 = \frac{1}{7} \\ H_a: \text{不全相同} \end{cases}$

$E_i = \frac{200}{7} = 28.57$

$\chi^2 = \sum \frac{(O_i - E_i)^2}{E_i} = 3.629$

$df = 7-1 = 6$

$\chi^2_{0.05,6} = 12.592$

∴ non-reject H_0

18. Fatal Accidents Accident data were analyzed to determine the numbers of fatal accidents for automobiles of three sizes. The data for 346 accidents are as follows:

	Small	Medium	Large	
Fatal	67	26	16	= 109
Not Fatal	128	63	46	= 237
	= 195	= 89	= 62	= 346

Do the data indicate that the frequency of fatal accidents is dependent on the size of automobiles? Test using a 5% significance level.

$E_{ij} = n \times P_{ri} \times P_{cj}$.

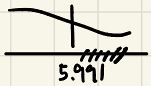
$\frac{109}{346} \times \frac{195}{346} \times 346 = 61.43$

61.43	28.04	19.53
133.57	60.96	42.47

$\begin{cases} H_0: & \text{Size \& fatal : independent} \\ H_a: & \text{Size \& fatal : dependent} \end{cases}$

$\chi^2 = \sum_{i=1}^{2}\sum_{j=1}^{3} \frac{(O-E)^2}{E} = 1.886 \qquad df = (r-1)(c-1) = 1 \times 2 = 2$

$\chi^2_{2,0.05} = 5.991$



$\therefore$ non-reject H_0

DATA SET DS1403

23. Telecommuting As an alternative to flextime, many companies allow employees to do some of their work at home. Individuals in a random sample of 300 workers were classified according to salary and number of workdays per week spent at home.

$n = 300$.

	Workdays at Home per Week			
Salary	Less Than One	At Least One, but Not All	All at Home	
Under \$50,000	38	16	14	= 68
\$50,000–\$74,999	54	26	12	= 92
\$75,000–\$99,999	35	22	9	= 66
Above \$100,000	33	29	12	= 94
	= 160	= 93	= 47	

a. Do the data present sufficient evidence to indicate that salary is dependent on the number of workdays spent at home? Test using $\alpha = .05$.

b. Use Table 5 in Appendix I to approximate the p-value for this test of hypothesis. Does the p-value confirm your conclusions from part a?

E_{ij}

36.267	21.080	10.653
49.067	28.520	14.413
35.200	20.460	10.340
39.467	22.940	11.593.

a. $\begin{cases} H_0: & \text{independent} \\ H_a: & \text{dependent.} \end{cases}$

$\alpha = 0.05$

$\chi^2_{STAT} = \frac{(O_{ij} - E_{ij})^2}{E_{ij}} \sim \chi^2_{(c-1)(r-1)}$.

$\chi^{2*} = \sum_{i=1}^{3}\sum_{j=1}^{4} \frac{(O_{ij} - E_{ij})^2}{E} = 6.447 \qquad df = (r-1)(c-1) = 6$

$\chi^2_{6,0.05} = 12.592$

$\therefore$ non-reject H_0

b. $p\text{-value} = P(\chi_{6,0.05} > 6.447)$
> 0.1

$0.1 > 0.05$

$\therefore$ non-reject H_0

$\therefore$ yes (same as part a)